Project Name: Green Lab Monitoring System

1. Current Limitations

While the current prototype successfully demonstrates smart laboratory resource monitoring, a few practical limitations are acknowledged:

- **Prototype Data Simulation:** For the current demo, sample sensor values and static dashboard data are used. In production, this will be connected to real IoT devices and a centralized cloud database such as Supabase PostgreSQL.
- **Basic Authentication:** The present system uses simple role-based demo login for faculty, students, and admin. Future deployment should integrate with the university's official login or SSO system.
- **Manual Sensor Mapping:** Lab sensor IDs and room mapping are currently configured manually. In real deployment, this should sync dynamically with the campus infrastructure database.
- **Limited Real-Time Devices:** The prototype assumes sensor inputs but does not yet connect with actual ESP32, smart meters, or gas flow devices.

2. Technical Limitations

- **Offline Support:** The current website requires internet connectivity for live monitoring. A future PWA version can support offline caching and sync.
- **Scalability:** Current logic is optimized for prototype-level data. For full campus deployment, the system would require load balancing, background queues, and realtime optimization.
- **Basic Anomaly Logic:** Leak and high-usage detection currently use threshold-based rules. More accurate AI-based anomaly detection can improve precision.

3. Future Enhancements (Roadmap)

To convert this into a production-grade smart campus solution, the following features are planned:

A. AI-Based Usage Prediction

Machine learning can be added to analyze historical water, gas, and electricity usage.

- **Objective:** predict unusual spikes, identify waste trends, and suggest preventive actions.

B. Mobile Application

A dedicated mobile app for faculty and maintenance teams.

- **Camera Upload:** upload photos of leaking taps or damaged equipment
- **QR Reporting:** scan lab QR code and instantly report issues
- **Push Notifications:** real-time alerts for leakage or after-hours usage

C. IoT Automation

Future versions will connect directly with:

- smart water flow sensors
- gas pressure sensors
- electricity smart meters
- occupancy sensors

This will allow **automatic ticket creation without manual reporting**.

D. Sustainability Intelligence

- monthly green lab certification
- department leaderboard
- carbon footprint reports
- predictive maintenance insights
- grant-wise cost optimization

4. Real-World Deployment Considerations

- **Data Privacy:** lab usage and faculty reports must be securely stored with encryption and role-based access.
- **Adoption Strategy:** gamification features like “Best Green Lab” and departmental sustainability rankings can increase engagement.
- **Maintenance Workflow:** automatic escalation rules should be added for unresolved critical leak alerts.
- **Campus Expansion:** the same system can scale to hostels, libraries, and smart classrooms.

5. One-Line Future Vision

The long-term vision is to transform every campus laboratory into a **smart, self-monitoring, sustainable research environment powered by IoT, analytics, and automation**.